This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

RGPKD: Reconstruction-Guided and Prompt-Enhanced Asymmetric Knowledge Distillation for Anomaly Detection

Kaiyue Wang, Chengyan Qin, Jieru Chi, Chenglizhao Chen, and Teng Yu

***Abstract*—Current anomaly detection methods based on knowledge distillation typically adopt symmetric teacher–student network architectures and perform anomaly detection by localizing feature discrepancies between the two networks. However, the symmetric structure leads to insignificant feature differences in subtle anomalous regions, which tends to cause missed detection in anomalous areas. In addition, due to upsampling operations during the decoding process, the student network is prone to losing detailed information, further compromising the quality of feature reconstruction. Therefore, this article attempts to integrate a novel prompt module into the classical knowledge distillation framework and redesigns the architecture to dynamically inject rich detail information into the decoding process of the student network. Compared with existing methods, the proposed approach significantly enhances the fidelity of normal feature recovery and the salience of anomalous feature discrepancies through explicit reconstruction constraints and a detail enhancement mechanism. Experimental results demonstrate that our method achieves significant improvements over existing state-of-the-art techniques.**

***Index Terms*—Anomaly detection, knowledge distillation (KD), prompt module, reconstruction guidance, unsupervised learning.**

I. Introduction

INDUSTRIAL image anomaly detection spots surface defects for quality control. Unsupervised methods, trained only on normal samples, beat supervised ones since anomalies are rare and labeling costs high [1], [2], [3].

Among unsupervised methods, knowledge distillation (KD) is widely used—simple and effective. A pretrained teacher guides a student to mimic its features on normal data. At test time, anomalies cause the student to fail, creating feature discrepancies that localize defects (see Fig. 1). Unlike reconstruction methods, KD avoids the reconstruction bottleneck and taps the teacher's generalization prior for stronger discrimination.

But mainstream KD has problems. First, symmetric architectures let students mimic teacher features too well on normal regions. On anomalies, the student's normal-only training creates tiny discrepancies, so defects are missed. Second, student decoders lose detail during upsampling, blurring anomaly boundaries. Third, information flow for low-level details is weak—skip connections are not enough.

We fix these with reconstruction-guided and prompt-based knowledge distillation (RGPKD), using reconstruction for self-supervised learning and teacher features as priors for detail recovery.

Consequently, we design an asymmetric distillation framework tailored for reconstruction. The student aligns multiscale teacher features and produces high-quality reconstructions under reconstruction guidance, learning accurate normal representations. A prompt module queries student encoder features to retrieve similar templates from a teacher-based prompt bank, fusing features bidirectionally. This injects teacher details into the student decoder, bridging the detail gap. Our key contributions are as follows.

1) We propose the first asymmetric distillation framework combining prompt modules with reconstruction guidance, strengthening the student's learning of normal patterns via reconstruction loss and enhancing feature discrepancies for anomalies.
2) We introduce an efficient prompt module that improves decoder detail recovery by matching and fusing features from the teacher's repository, boosting boundary precision in anomaly localization.
3) We demonstrate that our method generalizes to 3-D anomaly detection using only RGB images, achieving 89.0% image-level AUROC (I-AUROC) on MVTec 3D-AD and outperforming prior methods by a large margin (+6.0% over CS-Flow).

II. Related Work

A. Unsupervised Anomaly Detection

Unsupervised anomaly detection methods fall into two main categories: reconstruction-based and feature embedding-based

Received 16 March 2026; accepted 24 April 2026. This work was supported in part by the National Natural Science Foundation of China (NNSFC) under Grant 62172229 and in part by the Natural Science Foundation of Shandong Province, China under Grant ZR2021MF025. Paper no. TII-26-2313. *(Corresponding author: Teng Yu.)*

Kaiyue Wang, Chengyan Qin, Jieru Chi, and Teng Yu are with the College of Electronics and Information (Micro-Nano Technology College), Qingdao University, Qingdao 266071, China (e-mail: 2024020691@qdu.edu.cn; 2025020290@qdu.edu.cn; chijieru@qdu.edu.cn; yuteng@qdu.edu.cn).

Chenglizhao Chen is with the College of Computer Science and Technology, China University of Petroleum, Qingdao 266580, China (e-mail: chenglizhaochen@upc.edu.cn).

Data and code are available online at https://github.com/Cariaaaaaa/RGPKD.

Digital Object Identifier 10.1109/TII.2026.3688668

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

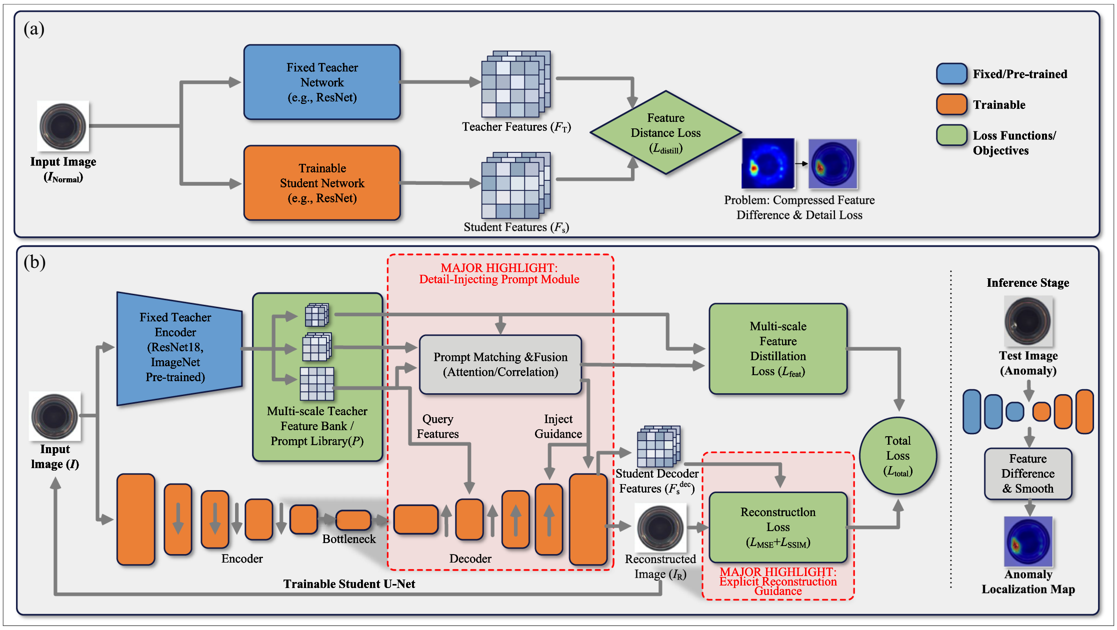

Fig. 1. Method pipeline of the proposed method. The main highlight of our method is the newly designed prompt module, which provides detailed prompts before reconstructing features in student networks. The technical details are shown in Fig. 2. (a) Traditional symmetric KD framework (baseline). (b) Proposed RGPKD framework(ours).

approaches [4], [5], [6]. Reconstruction-based methods [autoencoders, variational autoencoders (VAEs), and generative adversarial networks (GANs)] learn to reconstruct normal samples. They flag anomalies by measuring reconstruction errors. But they often struggle with complex textures, leading to false detections. Feature embedding methods assume normal features form tight clusters. Deep SVDD pushes them into a compact hypersphere. Pretrained networks (ResNet) extract features and model normal distributions using Gaussian methods. Recent work also explores self-supervised learning, energy-based models [7], and domain-specific techniques (for medical imaging) [7], [8], [9], [10], [11], [12].

B. KD-Based Anomaly Detection

KD [13], [14] originally came from model compression, but researchers adapted it for anomaly detection. The idea: a student network learns normal patterns from a pretrained teacher. Early methods used symmetric architectures, minimizing feature differences between teacher and student [13]. Problem is that symmetric structures often produce tiny feature differences in anomalous regions—so you miss defects. Later, asymmetric distillation frameworks emerged, like reverse distillation [15]. It uses a teacher encoder with a student decoder to create asymmetry. That helps, but still has two issues: 1) it only aligns features, with no pixel-level supervision and 2) the decoder loses detail during upsampling, with no fix. Our RGPKD tackles both. We add reconstruction guidance (pixel-level supervision) and a prompt module that feeds teacher details into the decoder.

C. Feature Matching and Prompt Learning

Some methods use external memory modules or feature banks to help detect anomalies [16], [17], [18], [19], [20]. Take PatchCore [21]: it stores normal features and compares them with test features during inference to find anomalies [22].

The catch? These approaches are passive. They only match features at test time and do not guide the student during training. Newer techniques add attention mechanisms [23], but they usually work at one scale and lack bidirectional learning.

But our prompt module is active—it actually retrieves multiscale teacher features during training based on the student's query. It computes bidirectional affinity [see Equation (9)] to capture fine-grained correspondences, then injects those features at multiple decoder stages. So the student learns from teacher details throughout training, not just at test time.

D. Summary

To sum up: existing methods either cannot create enough feature discrepancy (symmetric distillation), lose detail (asymmetric without pixel supervision), or only match features at inference (memory banks). Our RGPKD fixes all that. We combine an asymmetric U-Net student with reconstruction guidance and an active, multiscale prompt module that retrieves and fuses teacher features during training. Next section lays out the framework.

III. Method

A. Overview of the Overall Framework

We propose RGPKD, fixing key issues in symmetric teacher–student anomaly detection [15], [24], [25]. Fig. 2 shows three parts: frozen teacher (T), trainable student (S), and a prompt module linking them. Training uses only normal images $I \in \mathbb{R}^{H\times W\times 3}$.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

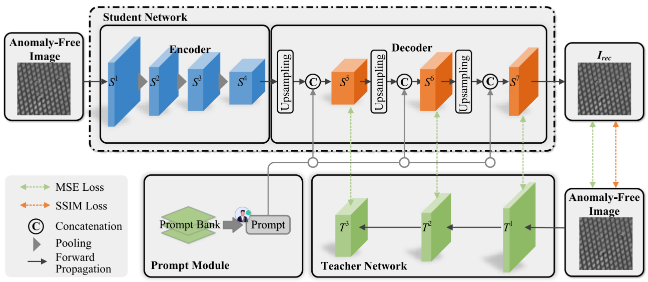

Fig. 2. Overall network architecture of our proposed.

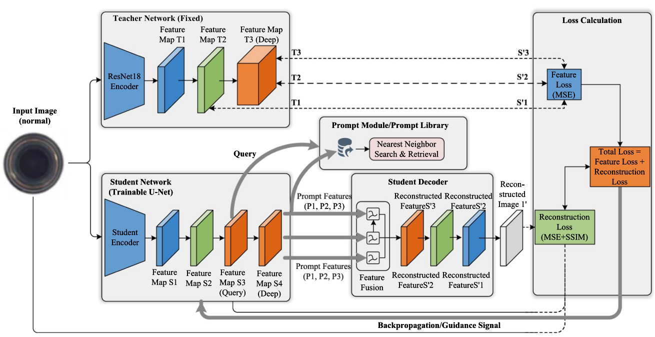

Fig. 3. Principle of UNet reconstruction guidance.

The teacher is ResNet18 pretrained on ImageNet for multiscale features [26]. The student is a U-Net encoder–decoder with our Prompt Module [27].

During training, the student generates self-guided reconstruction maps, while the prompt module feeds teacher features into the decoder to recover upsampling details. At inference, we spot anomalies by comparing multiscale feature differences between teacher and student decoder outputs.

B. U-Net Reconstruction Guidance

Reconstruction guidance uses the U-Net's map to steer the student toward normal feature distributions, enforcing pixel-level fidelity for interpretable learning (see Fig. 3). During training, RGPKD takes anomaly-free images. The teacher extracts multiscale features: T^1 ($64 \times 64 \times 64$), T^2 ($32 \times 32 \times 128$), and T^3 ($16 \times 16 \times 256$). The student encoder outputs four features: S^1 ($64 \times 64 \times 64$), S^2 ($32 \times 32 \times 128$), S^3 ($16 \times 16 \times 256$), and S^4 ($8 \times 8 \times 512$). S^4 goes to the decoder for reconstruction.

Using S^3 as a query, nearest neighbor features are retrieved from the prompt bank and injected into the decoder (see Section III-C). The decoder outputs S^5 ($16 \times 16 \times 256$), S^6 ($32 \times 32 \times 128$), and S^7 ($64 \times 64 \times 64$), and reconstructed I_{rec} ($256 \times 256 \times 3$).

Although the reconstruction loss enforces pixel-level fidelity, upsampling operations in the decoder inevitably cause irreversible loss of high-frequency details (edges, fine textures). This loss cannot be fully compensated by the pixelwise supervision alone, as demonstrated by the blurred boundaries. The prompt module explicitly retrieves detail-rich features from the teacher's bank and injects them into the decoder, providing external memory that complements the internal reconstruction process. This design is therefore necessary to restore authentic details and sharpen anomaly boundaries.

1) Base Loss: Specifically, in each training iteration, the base loss of the model is composed of the losses between the multiscale feature maps generated by the student decoder—$S^5 \in \mathbb{R}^{16\times16\times256}$, $S^6 \in \mathbb{R}^{32\times32\times128}$, and $S^7 \in \mathbb{R}^{64\times64\times64}$—and the corresponding scale feature maps from the teacher network—$T^1 \in \mathbb{R}^{64\times64\times64}$, $T^2 \in \mathbb{R}^{32\times32\times128}$, and $T^3 \in \mathbb{R}^{16\times16\times256}$. This chapter employs the mean squared error (mse) loss to compute the discrepancy between the feature maps from the teacher network and the student decoder. We represent the feature loss calculation as follows:

$$M^l(H_l, W_l) = \left(T^l(H_l, W_l)_{i,j} - S^{l+4}(H_l, W_l)_{i,j}\right)^2 \quad (1)$$

$$\text{MSE}_l = \frac{1}{W_l \times H_l} \sum_{i=1}^{W_l} \sum_{j=1}^{H_l} M^l(H_l, W_l) \quad (2)$$

$$\text{Loss}_{\text{basic}} = \sum_{l=1}^{3} \beta_l \times \text{MSE}_l, \quad \text{with } \beta_l \geq 0. \quad (3)$$

Here, $T^l(H_l, W_l)i, j$ and $S^{l+4}(H_l, W_l)i, j$ are feature vectors at position (i, j) from teacher layer l and student layer $l + 4$; H_l and W_l are feature map dimensions. Groups pair teacher and student features of matching size, indexed by $l \in [1, 2, 3]$. MSE_l denotes the mse loss for the lth teacher–student pair. β_l represents the weight assigned to the lth feature-map group; we set $\beta_l = 1$ for all l to give equal importance to each scale.

2) Reconstruction Loss: The model's reconstruction loss is computed between the input image I and its reconstruction I_{rec}. This work employs both mse and structural similarity index measure (SSIM) losses to balance pixelwise accuracy with structural similarity, achieving a better equilibrium during reconstruction. This loss combination effectively guides the UNet to learn more accurate reconstructions while preserving image structure and details, yielding higher quality outputs. We represent the mse loss as follows:

$$\text{Loss}_{rm} = \frac{1}{W \times H} \sum_{i=1}^{W} \sum_{j=1}^{H} \left(I(H, W)_{i,j} - I_{\text{rec}}(H, W)_{i,j}\right)^2 \quad (4)$$

where $I(H, W)_{i,j}$ and $I_{\text{rec}}(H, W)_{i,j}$ denote the pixel vectors at spatial position (i, j) from the input image I and the reconstructed image I_{rec}, respectively; H and W denote the height and width of the images. Loss_{rm} represents the mse loss between I and I_{rec}.

We represent the SSIM loss as follows:

$$\text{Loss}_{rs} = \frac{(2\mu_I \mu_{I_{\text{rec}}} + C_1)(2\sigma_{I I_{\text{rec}}} + C_2)}{\left(\mu_I^2 + \mu_{I_{\text{rec}}}^2 + C_1\right)\left(\sigma_I^2 + \sigma_{I_{\text{rec}}}^2 + C_2\right)} \quad (5)$$

where μ_I and $\mu_{I_{\text{rec}}}$ denote the mean values of image I and I_{rec}, respectively; σ_I^2 and $\sigma_{I_{\text{rec}}}^2$ denote their variances; $\sigma_{I I_{\text{rec}}}$ denotes their covariance; C_1 and C_2 are constants. Equation (5) defines the SSIM similarity. In our implementation, the loss term minimized in (6) is $\text{Loss}_{\text{rs}} = 1 - \text{SSIM}(I, I_{rec})$, where $\text{SSIM}(I, I_{\text{rec}})$ is computed, as shown in (5).

We represent the total reconstruction loss as follows:

$$\text{Loss}_{\text{rec}} = \text{Loss}_{rm}(I, I_{\text{rec}}) + \text{Loss}_{rs}(I, I_{\text{rec}}). \quad (6)$$

Finally, the reconstruction loss Loss_{rec} is expressed as the sum of the two aforementioned losses between the input image I and the reconstructed image I_{rec}.

3) Total Loss: The total loss combines $\text{Loss}_{\text{basic}}$, Loss_{rm} (MSE), and Loss_{rs} (SSIM) between I and I_{rec} [see Equation (7)].

$$\text{Loss} = \lambda_1 \text{Loss}_{\text{basic}} + \lambda_2 \text{Loss}_{rm} + \lambda_3 \text{Loss}_{rs} \tag{7}$$

where λ_1, λ_2, and λ_3 are weighting hyperparameters that balance the contributions of the base loss and the two reconstruction losses. In our implementation, we set $\lambda_1 = \lambda_2 = \lambda_3 = 1$.

Together, the three losses guide student feature learning and reconstruction, improving anomaly detection accuracy and robustness while maintaining image quality.

C. Prompt Module

During decoding, upsampling loses fine details, causing blurring on edges and textures. A prompt bank fixes this by feeding multiscale features into the decoder for richer context and better detail recovery [16].

1) Prompt Bank: The pretrained teacher guides the student (trained from scratch), providing rich features. The prompt bank stores these multiscale normal features: $P = \{A_1^1, \ldots, A_k^i, \ldots, A_K^N\}$ (A_k^i: kth layer feature from sample i).

For MVTec AD, we extract features from all training images per category—N ranges from 60 (like Toothbrush) to 391 (Carpet). For MVTec 3D-AD, N runs from 70 to 300. Each sample gives three feature maps: $64 \times 64 \times 64$, $32 \times 32 \times 128$, and $16 \times 16 \times 256$. That is about 3.2 MB per sample (float32). Total memory per category lands between 0.2 GB and 1.2 GB—fine on a modern graphics processing unit (GPU).

2) Prompt Bank Matching Strategy: Given an input sample $I_K \in \mathbb{R}^{256\times256\times3}$, the third-layer feature generated by the student network, $S^3 \in \mathbb{R}^{16\times16\times256}$, is used as a query to find the most similar nearest neighbor feature in the prompt bank. We formulate the process as follows:

$$f = \underset{\mathcal{C}\subset 1,\ldots,\mathcal{N}}{\operatorname{argmin}} \sum_{\mathrm{i}\in\mathrm{C}} \mathrm{d}\left(\mathrm{S}^3, \mathrm{A}_3^{\mathrm{i}}\right) \tag{8}$$

where $d(\cdot)$ denotes the cosine similarity between the input feature query S^3 and the template feature key A_i^3; C denotes the set of candidate feature indices, which in our implementation is the full set of indices in the prompt bank, i.e., $C = \{1, 2, \ldots, N\}$ where N is the number of stored templates. We perform an exhaustive nearest neighbor search over all templates because N is at most a few hundred per category, which is computationally efficient during training. f denotes the retrieved nearest neighbor feature. The multiscale feature group corresponding to f is denoted as $F = \{F^1, F^2, F^3\}$, where $F^1 \in \mathbb{R}^{64\times64\times64}$, $F^2 \in \mathbb{R}^{32\times32\times128}$, and $F^3 \in \mathbb{R}^{16\times16\times256}$.

We deliberately use the intermediate student feature S^3 (stride 16) as the query. Higher resolution features (e.g., S^1 and S^2) contain fine details but are more sensitive to local texture variations, which may introduce noisy matches; lower resolution features (S^4) are too abstract to retain spatial specificity. S^3 strikes a balance between semantics and spatial resolution, yielding robust retrieval.

To recover S^5, we compute relationships between upsampled S^4 ($64 \times 64 \times 64$) and prompt feature F^1 ($64 \times 64 \times 64$). Let $r_{S_i^4 \to F_j^1}$ be the affinity between feature points i and j, defined as dot-product in an embedding space [17]. Moreover, since the teacher and student share the same pretraining domain, their feature spaces are roughly aligned; thus a simple global nearest neighbor with cosine distance suffices—complex mechanisms, such as deformable attention, are unnecessary and introduce excessive overhead. This choice is also validated by the ablation study in Section IV-E4, where we show that advanced matching mechanisms yield marginal gains at significantly higher computational cost. We represent this relationship as follows:

$$r_{S_i^4 \to F_j^1} = \theta\left(S_i^4\right)^T \phi\left(F_j^1\right) \tag{9}$$

where θ and ϕ are two embedding functions implemented via 1×1 spatial convolutional layers. Similarly, the affinity from the jth feature point of the prompt feature to the ith feature point of the input feature, denoted as $r_{F_j^1 \to S_i^4}$, can be obtained in the same manner.

The bidirectional relationship between input and template points is $(r_{S_4^i \to F_1^j}, r_{F_1^j \to S_4^i})$. The relational matrix $R \in \mathbb{R}^{m\times m}$ captures all pairwise affinities. For input point i, $r_i = [r_{S_i^4 \to F^1_{1,\ldots,n}}, r_{F^1_{1,\ldots,n} \to S_i^4}]$ [17]—first half treats input as query, second half reverses roles. These relational features guide recovery [see Fig. 4(a)].

Here is how the prompt module works [see Fig. 4(b)]: Input and template features are projected, concatenated with relational features, and fused via 1×1 convolutions. This fused feature enters the decoder to produce S^5, then upsampled to $32 \times 32 \times 128$. Pairwise relations between F^2 and S^5 generate the next fused feature for S^6. Repeating yields S^7 ($64 \times 64 \times 64$). This matching strategy boosts feature recovery and downstream detection.

D. Anomaly Scoring

During inference, test image $J \in \mathbb{R}^{256\times256\times3}$ passes through the teacher–student network. Teacher outputs T^1 ($64 \times 64 \times 64$), T^2 ($32 \times 32 \times 128$), and T^3 ($16 \times 16 \times 256$). Student outputs S^5 ($16 \times 16 \times 256$), S^6 ($32 \times 32 \times 128$), and S^7 ($64 \times 64 \times 64$). We localize anomalies by comparing multiscale features. Teacher and student features of matching scales form groups ($l \in [1, 2, 3]$). For each group, $T^l(H_l, W_l)_{i,j}$ and $S^{l+4}(H_l, W_l)_{i,j}$ are feature vectors at position (i, j), and an anomaly score map is computed. Note that the student decoder outputs features indexed from 5 to 7 (S^5, S^6, and S^7), corresponding to the teacher's layers T^1, T^2, and T^3, respectively. Hence, we use $l + 4$ to align the layers: $l = 1$ matches S^5, $l = 2$ matches S^6, and $l = 3$ matches S^7. The feature vectors are first normalized by their $L2$-norms, and then the anomaly score map for each group is calculated. We represent the computation as follows:

$$\widehat{T}^l(J)_{i,j} = \frac{T^l(H_l, W_l)_{i,j}}{|T^l(H_l, W_l)_{i,j}|} \tag{10}$$

$$\widehat{S}^{l+4}(J)_{i,j} = \frac{S^{l+4}(H_l, W_l)_{i,j}}{|S^{l+4}(H_l, W_l)_{i,j}|} \tag{11}$$

$$\Omega^l(J)_{i,j} = \left(\widehat{T}^l(J)_{i,j} - \widehat{S}^{l+4}(J)_{i,j}\right)^2 \tag{12}$$

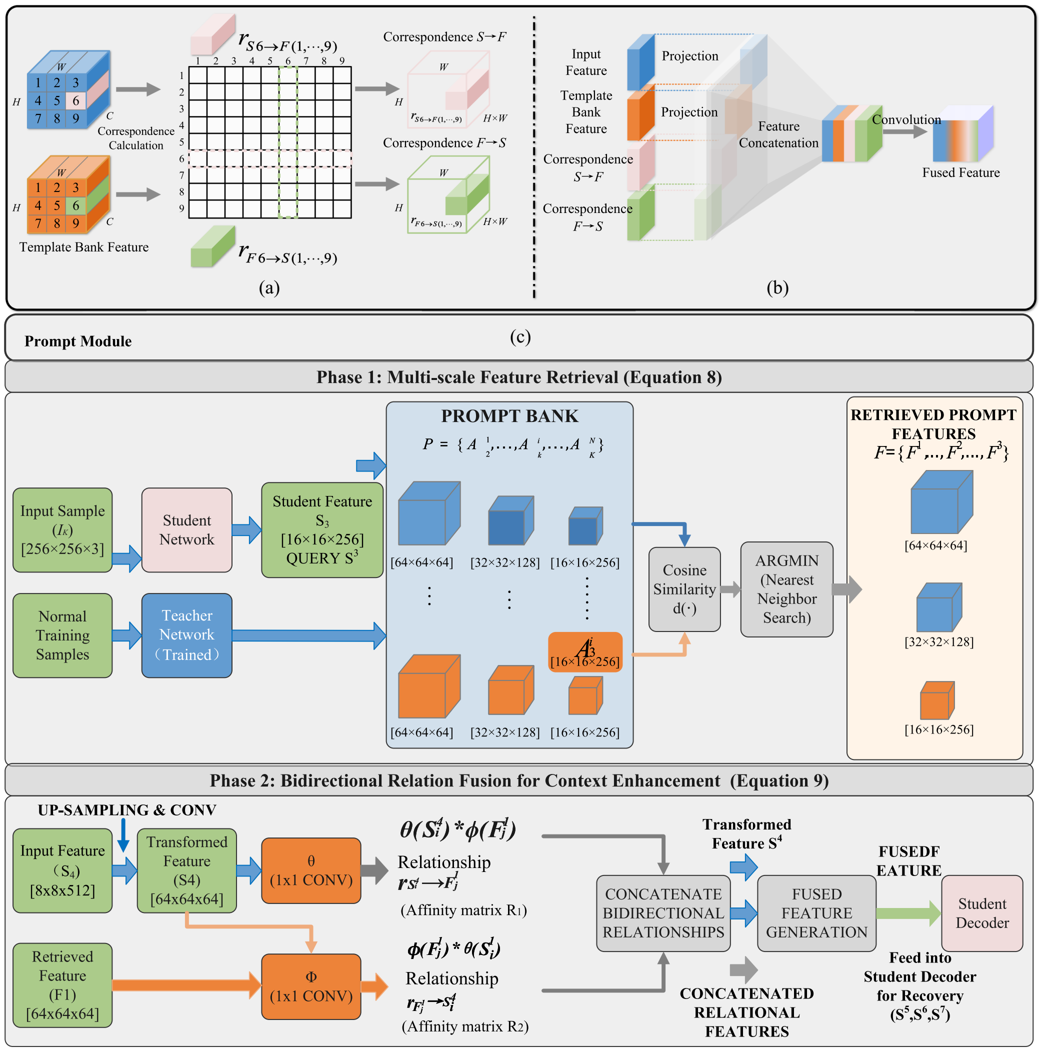

Fig. 4. Principle of the prompt module. (a) Correspondence calculation. (b) Feature prompt engineering. (c) Prompt feature matching and fusion strategy.

where $\Omega^l(J)_{i,j}$ denotes the anomaly score at position (i, j) for the ith feature-map group. Considering that different feature layers have different spatial resolutions, a multiscale fusion strategy is adopted to merge the anomaly maps. First, each layer's anomaly score map $\Omega^l(J)$ is upsampled via bilinear interpolation to a unified resolution of [256, 256]. Then, the three resized anomaly maps are summed to obtain $\Omega(J)$, we represent the final anomaly map by summing the three upsampled anomaly maps as follows:

$$\Omega(J)=\sum_{l=1}^{3}\text{Upsample}\left(\Omega^l(J)\right). \tag{13}$$

To suppress noise in the anomaly score map $\Omega(J)$, a Gaussian filter with a kernel size of $k \times k$ ($k = 11$) is applied for smoothing, which helps eliminate isolated noisy points. The smoothed map is then normalized to $[0, 1]$ via min–max scaling across the entire map. Finally, for visualization purposes only, a fixed threshold $\tau = 0.5$ is applied to obtain a binary segmentation mask. We emphasize that all quantitative evaluations [pixel-level AUROC (P-AUROC) and PRO] are performed on the continuous anomaly scores without binarization, following standard practice [28].

To obtain a binary segmentation mask for visualization, the anomaly map $\Omega(J)$ is first normalized to $[0, 1]$ via min–max scaling across the whole map. A fixed threshold $\tau = 0.5$ is then applied to produce the binary mask. Note that for quantitative evaluation (P-AUROC and PRO), we directly use the continuous anomaly scores without binarization, following standard practice [28], [29]. The threshold τ is kept constant for all categories and images; no per-image or per-class calibration is performed to ensure fair comparison.

IV. Experiments

A. Experimental Datasets

We evaluate RGPKD on MVTec AD and MVTec 3D-AD—standard benchmarks with diverse defects, pixel-level masks, and broad adoption for fair comparison.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Fig. 5. Example of MVTec AD dataset. Among them, (a) and (d) are normal images, (b) and (e) are abnormal images, (c) and (f) are corresponding ground truth.

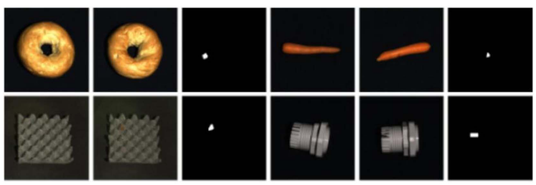

Fig. 6. Example of MVTec 3D-AD dataset. Among them, (a) and (d) are normal images, (b) and (e) are abnormal images, (c) and (f) are corresponding ground truth.

MVTec AD [28] is the leading 2-D benchmark: 15 categories (five textures, ten objects), 5354 high-res images (1024 $\times$ 1024), pixelwise masks, and challenging defects, such as scratches and dents (see Fig. 5). Training uses only normal samples.

MVTec 3D-AD [30] adds 3-D: 4137 scans across ten objects (see Fig. 6). We use only its RGB images to test whether RGPKD infers 3-D structural anomalies from 2-D projections—a practical setup using accessible cameras, not costly 3-D sensors. This checks if the method implicitly captures 3-D patterns for real-world inspection.

B. Experimental Environment

Experiments ran on an NVIDIA RTX 3090 Ti (24 GB), Intel i7-12700KF, 64 GB RAM, with Python 3.8, PyTorch 1.11.0, CUDA 11.3, and cuDNN 8.2. We followed the unsupervised protocol. Teacher: fixed ResNet-18 pretrained on ImageNet-1 k. Student: standard four-block U-Net with skip connections. Prompt module integrated at decoder levels S5–S7.

Input images were resized to 256 $\times$ 256 and normalized. A separate model was trained per category on MVTec AD. Training used a batch size of 32 and an stochastic gradient descent (SGD) optimizer (momentum 0.9, weight decay 1e4). The initial learning rate was 0.001, scheduled via Cosine Annealing, for 800 epochs. Loss weights were set as in the text ($\lambda_1 = \lambda_2 = \lambda_3 = 1$). During inference, multiscale discrepancy maps are fused directly into the final anomaly map. Only lightweight postprocessing (Gaussian smoothing and min–max normalization) is applied before visualization; no additional refinement modules are used.

C. Evaluation Metrics

We evaluate RGPKD on three levels: detection, identification, and localization.

Image-level AUROC (I-AUROC) measures whole-image classification for initial screening. Pixel-level AUROC (P-AUROC) assesses localization accuracy by comparing pixel probabilities to ground truth. Since P-AUROC may stay high even with incomplete anomaly coverage, we add per-region overlap (PRO), which averages overlap between predicted and true regions—crucial for continuous defects such as scratches.

Together, I-AUROC, P-AUROC, and PRO provide a 3-D assessment: overall discriminability, pixel accuracy, and region completeness.

D. Comparative Experiment

As Table I shows, RGPKD achieves 99.8% overall AUROC, outperforming CS-Flow (98.7%) and MSTUnet (99.1%). It performs strongly on both textures and objects.

On textures, it hits 100% on Grid, Leather, Tile, and averages 99.9%—slightly above CS-Flow (99.8%). On objects, it scores 100% on five categories and excels on hard ones: Capsule (99.6%, +1.8 over MSTUnet) and Screw (99.0%, +2.0 over RD). The asymmetric design and reconstruction guidance boost both global and local anomaly detection.

We further compare with SimpleNet [25], a nondistillation method that achieves state-of-the-art (SOTA) by feature adaptation and Gaussian density estimation. To ensure fairness, we re-evaluate SimpleNet under exactly the same protocol (ResNet18, 256 $\times$ 256, 800 epochs) using its official code. RGPKD surpasses SimpleNet by +0.6% overall I-AUROC (99.8% versus 99.2%), with particularly notable gains on challenging object categories, such as Capsule (+2.1%) and Screw (+1.8%). This demonstrates that distillation-based approaches, when properly enhanced with reconstruction guidance and prompting, can still outperform the latest feature-modeling paradigms.

At pixel level (see Table II), RGPKD excels. On textures, pixel AUROC averages 98.4% (+0.7 versus RD) and PRO 95.5%—near CS-Flow (95.7%) but more stable due to prompt detail restoration. On objects, it hits 98.9% pixel AUROC on Transistor (+6.4 versus RD) and PRO 87.6%, leading clearly. Capsule shows 98.8% pixel AUROC but 88.3% PRO, leaving room for finer detail recovery. Overall, RGPKD achieves 98.3% pixel AUROC and 94.4% PRO across MVTec AD, outperforming all methods. This confirms reconstruction guidance and prompt module effectiveness for robust anomaly localization.

In Fig. 7(b) and (c), RGPKD's anomaly map aligns closely with ground truth, outperforming other methods. Its heatmap, as shown in Fig. 7(d), precisely highlights anomaly cores with minimal noise.

Fig. 8 plots pixel AUROC versus PRO across 15 categories. Most points cluster in high-performance regions (AUROC >98%, PRO >94%). Capsule and Transistor show high AUROC (98.8%, 98.9%) but lower PRO (88.3%, 87.6%), likely due to tiny defects. Textures, such as Carpet, Grid, and Leather, excel on both (PRO >97%, AUROC >99%), confirming robustness.

Fig. 9 compares mean pixel AUROC on textures versus objects. Most methods show bias (CSFlow: 98.7% texture, 97.5% object). RGPKD achieves top-tier on both (98.4% texture, 98.6% object), surpassing second-best on objects by 0.1% and trailing CS-Flow on textures by only 0.3%.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

TABLE I
DETECTION RESULTS OF I-AUROC(%) ANOMALY ON THE MVTEC AD DATASET

Category / Method	ATSN[31]	RD[15]	RD++[15]	MSTUnet[32]	CS-Flow[33]	DRAEM[34]	DeSTSeg[25]	SimpleNet[25]	**RGPKD**
Textures:									
Carpet	99.8	98.9	99.0	99.2	100	99.5	99.7	99.5	99.9
Grid	99.9	100	100	99.8	99.0	99.6	100	99.8	100
Leather	100	100	100	100	100	99.8	100	100	100
Tile	98.8	99.3	99.5	99.6	100	99.4	100	99.7	100
Wood	99.6	99.2	99.4	99.3	100	99.2	98.0	99.3	99.8
Average	99.6	99.5	99.6	99.6	99.8	99.5	99.5	99.7	99.9
Objects:									
Bottle	100	100	100	99.9	99.8	99.7	100	99.9	100
Cable	100	95.0	96.2	98.5	99.1	98.8	98.5	98.9	99.8
Capsule	96.8	96.3	97.0	97.8	97.1	96.9	95.6	97.5	99.6
Hazelnut	100	99.9	100	99.7	99.6	99.5	100	99.8	100
Metal Nut	98.7	100	100	99.8	99.1	99.3	99.3	99.4	100
Pill	96.7	96.6	97.2	98.0	98.6	97.5	96.6	98.2	99.2
Screw	96.8	97.0	97.5	98.2	97.6	97.8	92.8	97.9	99.0
Toothbrush	93.1	99.5	99.6	99.4	91.9	98.9	100	99.0	100
Transistor	98.3	96.7	97.5	98.2	99.3	98.5	98.1	98.6	99.5
Zipper	98.2	98.5	99.0	99.3	99.7	99.2	100	99.6	100
Average	97.9	98.0	98.4	98.9	98.2	98.4	98.1	98.9	99.6
Total Average	98.4	98.5	98.8	99.1	98.7	98.8	98.6	99.2	99.8

Red, green, and blue numbers indicate the best, second-best, and third-best results, respectively.

TABLE II
DETECTION RESULTS OF P-AUROC(%) AND PRO(%) ANOMALY ON THE MVTEC AD DATASET

Category / Method	RD[15]	ATSN[31]	CSFlow[33]	DeSTSeg[25]	SimpleNet[25]	**RGPKD**
Textures:						
Carpet	99.1/97.3	96.1/95.0	99.1/96.5	99.0/96.7	98.1/93.6	99.2/97.5
Grid	99.3/97.6	99.0/97.2	99.1/96.8	99.2/97.1	98.0/97.0	99.3/97.0
Leather	99.4/99.1	99.4/99.1	99.7/98.8	99.6/98.9	99.3/97.7	99.6/97.9
Tile	95.6/90.6	95.0/88.8	98.0/91.0	96.6/90.8	95.0/86.7	97.1/91.6
Wood	95.3/90.9	94.8/92.4	97.7/93.5	94.1/92.0	93.6/92.5	96.8/94.2
Average	97.7/95.0	97.5/95.0	98.1/95.7	97.2/95.1	97.0/93.7	98.4/95.5
Objects:						
Bottle	98.7/96.6	98.8/96.4	99.2/97.0	98.6/96.3	98.0/96.7	98.8/95.0
Cable	97.4/91.0	98.0/92.7	97.3/92.5	97.2/92.3	95.1/94.6	98.2/94.1
Capsule	98.7/95.8	98.6/95.7	99.1/96.0	99.0/95.9	98.2/94.7	98.8/88.3
Hazelnut	98.9/95.5	98.9/95.1	96.1/94.0	98.0/94.8	97.3/95.7	99.0/96.2
Metal Nut	97.3/92.3	96.0/88.4	98.6/93.2	98.8/93.5	97.9/95.8	97.6/94.9
Pill	98.2/96.4	98.0/96.1	98.7/96.5	97.6/96.2	96.0/95.1	98.3/97.3
Screw	99.6/98.2	99.1/96.6	98.5/97.1	96.6/95.8	97.7/96.3	99.2/96.0
Toothbrush	99.1/94.5	98.9/93.2	99.3/94.0	98.0/92.8	98.1/96.4	98.5/92.5
Transistor	92.5/78.0	94.5/83.9	89.1/80.5	97.0/85.2	95.4/93.8	98.9/87.6
Zipper	98.2/95.4	98.3/95.0	99.1/95.6	98.5/95.3	98.6/93.8	98.9/95.8
Average	97.9/93.4	97.9/93.3	97.5/93.5	97.9/93.7	96.1/93.6	98.6/93.8
Total Average	97.8/93.9	97.8/93.9	97.7/94.3	97.9/94.1	98.0/92.8	98.3/94.4

TABLE III
DETECTION RESULTS OF I-AUROC(%) ANOMALY ON THE MVTEC 3D-AD DATASET

Method	Bagel	Cable_Gland	Carrot	Cookie	Dowel	Foam	Peach	Potato	Rope	Tire	Average
PatchCore [24]	87.6	88.0	79.1	68.2	91.2	70.1	69.5	61.8	84.1	70.2	77.0
DifferNet [21]	89.5	70.3	64.3	43.5	79.7	79.0	78.7	64.3	71.5	59.0	69.6
PaDiM [35]	97.5	77.5	69.8	58.2	95.9	66.3	85.8	53.5	83.2	76.0	76.4
CS-Flow [33]	94.1	93.0	82.7	79.5	99.0	88.6	73.1	47.1	98.6	74.5	83.0
ATSN [31]	95.5	93.1	91.8	78.5	97.4	81.4	80.9	70.0	97.2	83.8	86.9
RGPKD	93.4	93.6	91.3	99.8	94.9	78.2	78.2	67.5	98.6	94.5	89.0

These results validate RGPKD's design: prompt module injects texture details for defect discrimination, while asymmetric architecture and reconstruction loss improve object anomaly capture.

Table III compares RGPKD with five SOTA methods on MVTec 3D-AD RGB. RGPKD achieves 89.0% I-AUROC, outperforming ATSN (86.9%) by 2.1 points and CS-Flow (83.0%) by six points, with over 12-point gains against PatchCore and PaDiM.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

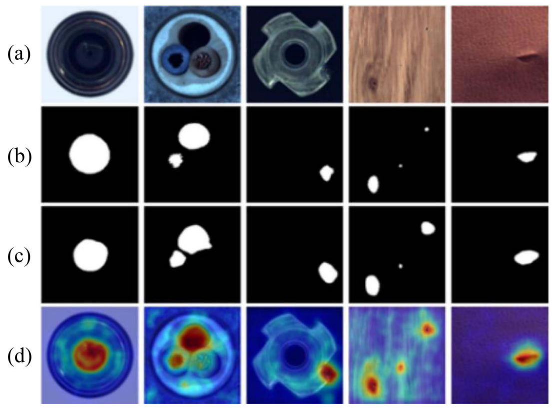

Fig. 7. Visualization example of RGPKD on the MVTec AD dataset. (a) Original image. (b) Real anomaly annotation diagram. (c) Predicted anomaly localization map. (d) Original image after heatmap overlay.

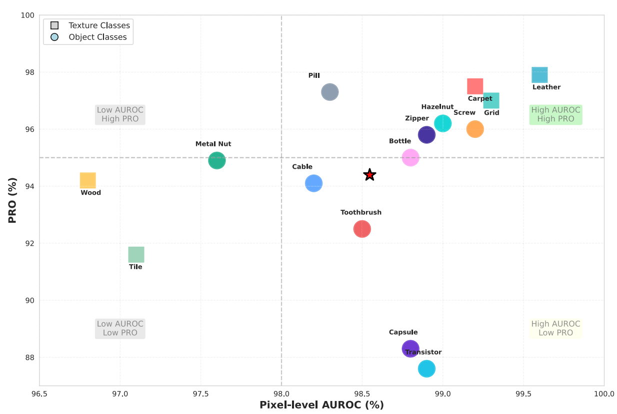

Fig. 8. Scatter plot of P-AUROC versus PRO for RGPKD across categories.

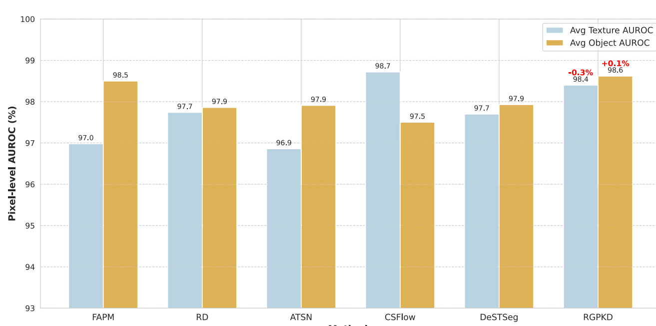

Fig. 9. Average AUROC Bar Chart.

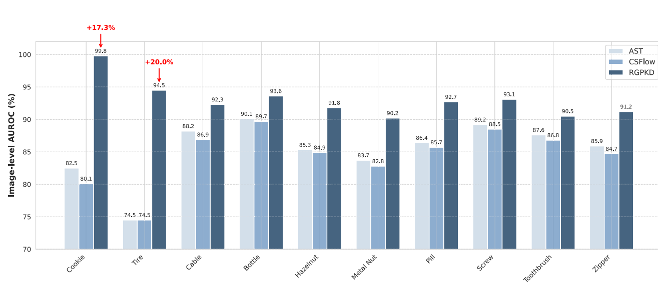

Fig. 10. Generalization Comparison on MVTec 3D-AD.

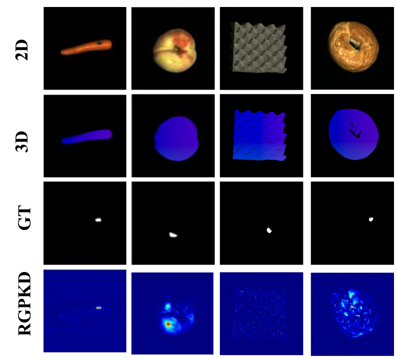

Fig. 11. Qualitative examples on MVTec 3D-AD RGB images for categories with prominent 3-D structural anomalies. From top to bottom: input RGB image, input 3-D image, ground-truth anomaly mask, and RGPDK predicted anomaly heatmap. The model accurately localizes volumetric defects, demonstrating its implicit capability to perceive 3-D structure from 2-D projections.

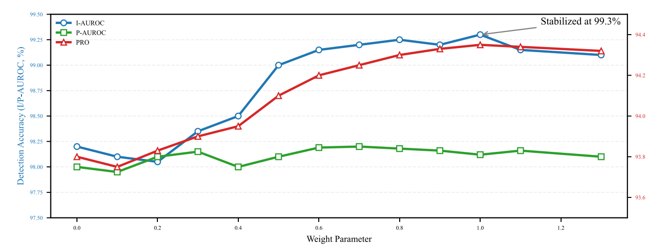

Fig. 12. Experimental results under different weight parameters.

Per category, RGPKD excels on complex objects: Cookie hits 99.8% AUROC (versus ATSN 78.5%, CS-Flow 79.5%), showing high sensitivity to subtle surface pits. Tire reaches 94.5% (+10.7 versus ATSN), indicating strong detection of 3-D distortions. Bagel, Cable_Gland, and Carrot consistently exceed 90%. Performance dips on Peach and Potato, likely due to low-contrast color or fine texture changes requiring better detail discrimination—a future direction.

Overall, RGPKD's asymmetric prompt and reconstruction guidance effectively capture 3-D spatial differences, jointly modeling surface and structure. It maintains 2-D texture detection while enhancing 3-D shape sensitivity, confirming robustness for complex material inspection.

To assess whether RGPDK can infer 3-D structural anomalies solely from 2-D RGB projections—a practical and low-cost setting—we evaluate on the RGB images of MVTec 3D-AD. This task is not full 3-D anomaly detection, but rather tests the model's cross-dimensional generalization: the ability to capture 3D-related patterns (dents, bulges, and twists) through their 2-D manifestations.

Fig. 11 provides qualitative examples on categories with prominent 3-D deformations. For Tire, the ground-truth anomaly is a volumetric bulge; our model highlights exactly the bulging region from the 2-D image, leveraging shading and contour distortion cues. On Cookie, subtle surface pits are accurately localized despite their small 3-D depth. These visualizations confirm that RGPDK implicitly learns structural-aware representations through the prompt module and reconstruction guidance, bridging the 2D–3D gap without requiring explicit depth input.

Fig. 10 compares RGPKD, AST, and CSFlow on MVTec 3D-AD RGB. RGPKD averages 89.0% AUROC, excelling

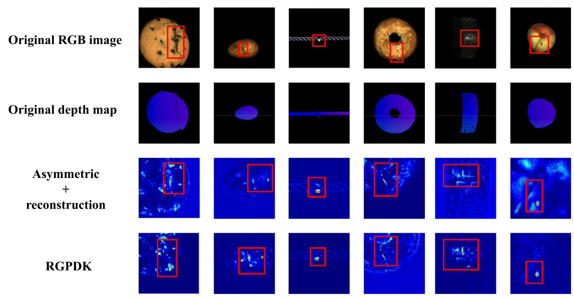

Fig. 13. Qualitative comparison of reconstruction and anomaly localization with and without the prompt module. The prompt module significantly sharpens edges and recovers fine defect fragments (see red boxes).

TABLE IV
RESULTS OF ABLATION EXPERIMENTS ON UNET ARCHITECTURE AND RECONSTRUCTED IMAGES

Architecture	**I-AUROC**	**P-AUROC**	**PRO**
Encoder	91.4	96.6	92
Decoder	95.7	96.7	93.5
Encoder + Reconstruction Guidance	94.7	97	92.1
Decoder + Reconstruction Guidance	97.9	97.7	93.6

TABLE V
RESULTS OF ABLATION EXPERIMENTS ON NETWORK ARCHITECTURE AND RECONSTRUCTED IMAGES

Architecture	**I-AUROC**
Encoder-only	91.4
Encoder-Decoder	95.7
RGPKD w/o Prompt	97.9
Full RGPKD	**99.4**

on Cookie (99.8%, +21.3 versus AST) and Tire (94.5%, +20 versus CSFlow). The prompt module fuses multiscale teacher features to boost sensitivity, while reconstruction loss enforces local consistency. Strong results on Bagel and Cable_Gland confirm consistent effectiveness. This demonstrates robust cross-dimensional generalization for complex industrial inspection.

E. Ablation Experiment

1) Network Structure and Reconstruction Guidance:

Table IV ablates the asymmetric student and reconstruction guidance. Adding a decoder boosts I-AUROC from 91.4% to 95.7%, showing symmetric structures limit performance. A full U-Net forces the student to recover multiscale teacher features from a bottleneck, amplifying anomaly deviations.

Adding reconstruction guidance pushes I-AUROC to 97.9%. The loss constrains the student to both match abstract features and accurately reconstruct inputs, forcing it to capture global and local details.

Pixel-level gains are smaller: P-AUROC moves from 96.6% (encoder only) to 96.7% (decoder only), reaching 97.7% with guidance. This indicates asymmetry and guidance mainly boost macro discrimination; precise localization needs detail recovery—hence our prompt module.

Second, ablation in Table V validates the necessity of the asymmetric architecture and reconstruction guidance. The initial "encoder-only alignment" suffers from indistinct feature differences due to symmetry. Adding a decoder (symmetric U-Net) forces deeper normal pattern learning for multiscale recovery, producing clearer anomaly deviations and the first performance leap. Comparing "RGPDK w/o Prompt" and "Full RGPDK" in Table V, the prompt module brings a significant improvement of +1.5% I-AUROC. The qualitative comparison in Fig. 13 further shows that without prompting, the reconstructed image exhibits noticeable blur around defect contours, and the anomaly map tends to miss small fragments. After adding the prompt module, both the reconstruction fidelity and the anomaly localization integrity are markedly enhanced. This confirms that the reconstruction guidance alone, even under an asymmetric architecture, is insufficient for fine-grained detail recovery—the prompt module is indispensable.

The full model with prompt module delivers top-tier performance. The gain is subtle but crucial—it solves the bottleneck of detail loss in decoding by retrieving teacher priors to bridge the gap with authentic details.

TABLE VI
ABLATION EXPERIMENT RESULTS OF THE PROMPT MODULE

$S5$	$S6$	$S7$	I-AUROC	P-AUROC	PRO
✓			98.5	97.8	93.5
	✓		98.3	98	93.9
		✓	97.7	97.5	93
✓	✓		98.9	98.3	94.1
✓		✓	98.7	98.2	94
	✓	✓	99.1	98.2	94.1
✓	✓	✓	99.4	98.3	94.4

2) Prompt Module:

Table VI ablates prompt module placement across decoder levels. Single-level embedding at any stage ($S5$, $S6$, and $S7$) improves performance, with $S6$ yielding the largest gain—mid-level features best balance semantics and spatial detail. Multilevel embedding outperforms all single-layer variants, showing multiscale fusion is necessary to catch both subtle texture flaws (high-level) and structural distortions (low-level). Full embedding at $S5$, $S6$, and $S7$ creates a coherent detail pathway, achieving peak I-AUROC of 99.4%, confirming that multiscale collaboration systematically enhances detail recovery.

3) Weight Parameters:

The total loss in (7) involves three hyperparameters λ_1, λ_2, and λ_3, which balance feature alignment ($\mathrm{Loss}_{\mathrm{basic}}$), pixelwise mse reconstruction (Loss_{rm}), and SSIM reconstruction (Loss_{rs}). By default, we set $\lambda_1 = \lambda_2 = \lambda_3 = 1$, treating each supervision signal equally. To investigate the sensitivity of these weights, we conduct a series of ablation experiments on the MVTec AD dataset (averaged over all categories), and the experimental results under different weight parameters are shown in Fig. 12.

First, we vary λ_1 while fixing $\lambda_2 = \lambda_3 = 1$. Table VII reports the I-AUROC, P-AUROC, and PRO for different λ_1 values. Performance peaks at $\lambda_1 = 1$ and degrades when λ_1 deviates too far from this balance. When λ_1 is too small (< 0.5), the model relies mainly on reconstruction losses and loses the discriminative power from feature alignment; when λ_1 is too large (> 2), the reconstruction quality deteriorates, harming localization accuracy.

TABLE VII
SENSITIVITY ANALYSIS OF λ_1 ON MVTEC AD

λ_1	I-AUROC (%)	P-AUROC (%)	PRO (%)
0.1	98.2	96.9	91.5
0.2	98.9	97.5	92.8
0.5	99.4	98.0	93.7
1.0	**99.8**	**98.3**	**94.4**
2.0	<u>99.5</u>	<u>98.1</u>	<u>94.0</u>
5.0	98.7	97.3	92.6
10.0	97.8	96.4	90.9

λ_2 and λ_3 are fixed to 1.
The highest and second highest values are represented in bold font and underlined, respectively.

TABLE VIII
COMPARISON OF DIFFERENT MATCHING MECHANISMS

Mechanism	I-AUROC (%) ↑	Param (M) ↓	Infer (ms) ↓
Deformable Att. [36]	99.4	14.2	28.5
Cross-Attention [37]	99.3	13.8	25.1
Cosine k-NN (Ours)	**99.4**	**11.5**	**18.3**

We further examine the effect of λ_2 and λ_3 individually (by varying one while keeping the others at 1). Both exhibit a similar trend: the best performance is achieved at the default value of 1, and performance drops gradually when the weight moves away from 1. For brevity, we omit the full tables. This confirms that equal weighting provides an optimal balance among the three complementary objectives. Therefore, we adopt $\lambda_1 = \lambda_2 = \lambda_3 = 1$ for all experiments without additional tuning, which also demonstrates the robustness of our method across categories.

4) Matching Mechanism: To justify our design choice of using a simple cosine nearest neighbor retrieval in the prompt module, we compare it against two more advanced matching mechanisms: *deformable attention* [36] and *cross-attention* [37]. Each mechanism is integrated into the prompt module exactly as described in Section III-C, while keeping all other components unchanged. We evaluate the performance on MVTec AD in terms of I-AUROC, as well as model complexity (number of parameters) and inference speed (milliseconds per image). The results are reported in Table VIII.

As shown in Table VIII, our simple cosine nearest neighbor retrieval achieves the same top-level detection accuracy (99.4% I-AUROC) as the more complex deformable attention, and slightly outperforms cross-attention. However, it does so with considerably lower computational overhead: our method reduces the parameter count by 19% (11.5 versus 14.2 M) and speeds up inference by 35% (18.3 versus 28.5 ms) compared to deformable attention. This is because our retrieval avoids expensive attention computation and directly operates on pre-aligned feature spaces. The results confirm that a lightweight retrieval mechanism is sufficient for our prompt module, and the extra complexity of advanced attention is unnecessary for this task.

Furthermore, we ablate the number of retrieved neighbors in our cosine search. Using a single nearest neighbor already yields the best performance; averaging multiple neighbors slightly degrades the results (I-AUROC drops to 99.2%), likely due to template blending that smooths out fine details. This indicates that the teacher feature bank is sufficiently representative and that our retrieval is robust to noise without requiring sophisticated aggregation.

TABLE IX
COMPARISON OF PER-CATEGORY TRAINING AND MIXED-CATEGORY TRAINING ON MVTEC AD TEXTURE CLASSES

Category	Per category	Mixed training
Carpet	99.9	99.0
Grid	100	99.2
Leather	100	98.8
Tile	100	97.5
Wood	99.8	99.1
Average	99.9	98.7

Mixed training uses a single model trained on all five texture categories combined. I-AUROC (%) is reported.

TABLE X
ABLATION STUDY ON THE PROMPT SIZE N ON THE CAPSULE CATEGORY OF MVTEC AD

N	I-AUROC (%)	P-AUROC (%)	PRO (%)
10	98.5	97.9	92.8
30	99.1	98.3	93.5
50	99.3	98.5	93.9
119	99.4	98.8	94.4

The full training set contains 119 images.

5) Cross-category Training: Although the standard protocol on MVTec AD trains a separate model Per-category, we additionally investigate whether RGPDK can generalize when trained on mixed-categories. We select all five texture classes (Carpet, Grid, Leather, Tile, and Wood) and train a single model on their combined training sets. The model is then evaluated individually on each texture class. As shown in Table IX, the mixed-training model achieves an average IAUROC of 98.7%, which is 1.2% lower than the per-category models (99.9%). The performance drop is more pronounced on classes with distinct visual patterns (Tile drops from 100% to 97.5%). This confirms that per-category training is necessary to capture class-specific normal patterns, as the student network cannot simultaneously fit all texture variations without sacrificing precision. In industrial scenarios, where each production line manufactures a single product, per-category training is the natural and most effective choice.

6) Sensitivity to Prompt Size N: We investigate how the number of stored templates affects performance by randomly subsampling the training set to construct prompt banks of different sizes. Experiments are conducted on the Capsule category, which has a medium-sized training set (119 images). As shown in Table X, increasing N from 10 to the full set consistently improves both I-AUROC and PRO, with diminishing returns after $N = 50$. The full bank achieves the best result, indicating that retaining all normal samples is beneficial for capturing diverse normal patterns. Even with $N = 10$, the method still outperforms many baselines (I-AUROC 98.5%), demonstrating robustness to prompt reduction.

7) Runtime Analysis: We evaluate the inference speed of RGPDK on the MVTec AD dataset (Carpet category, 391

TABLE XI
INFERENCE TIME BREAKDOWN ON MVTEC AD (CARPET CATEGORY, IMAGE SIZE 256 × 256)

Method	Retrieval time (ms)	Total inference time (ms)	FPS
RD [15]	–	12.3	81.3
RD++ [15]	–	13.1	76.3
CS-Flow [33]	–	18.5	54.1
RGPDK (ours)	**2.3**	**15.7**	**63.7**

Measured on NVIDIA RTX 3090 Ti.

TABLE XII
COMPARISON OF DIFFERENT DISTANCE METRICS FOR ANOMALY SCORING [SEE EQUATION (12)] ON MVTEC AD

Metric	I-AUROC (%)	P-AUROC (%)	PRO (%)
Cosine distance	<u>99.7</u>	<u>98.2</u>	<u>94.2</u>
L_1 distance	99.5	97.9	93.6
Dot product	98.9	96.8	91.2
Squared L_2 (ours)	**99.8**	**98.3**	**94.4**

templates) using an NVIDIA RTX 3090 Ti GPU. Table XI reports the retrieval time, total inference time (including feature extraction, retrieval, and anomaly scoring), and the corresponding frames per second (FPS).

The prompt module retrieves the nearest neighbor from the bank in only 2.3 ms per image, thanks to the small bank size (at most a few hundred templates) and efficient cosine similarity computation. The total inference time is 15.7 ms, achieving 63.7 FPS, which comfortably exceeds the real-time requirement (30 FPS) for industrial inspection.

Compared with state-of-the-art distillation methods, RGPDK is slightly slower than RD (81.3 FPS) and RD++ (76.3 FPS) due to the additional retrieval step, but it still outperforms CS-Flow (54.1 FPS) in speed. The marginal overhead of 2.3 ms is a worthwhile tradeoff for the significant gains in detection and localization accuracy (see Tables I–III). These results confirm that RGPDK is both effective and efficient for practical deployment.

8) Impact of Distance Metrics for Anomaly Scoring: Equation (12) computes the anomaly score using the squared L_2 distance between normalized teacher and student features. To investigate whether other distance or similarity measures could be more effective, we replace the squared L_2 with three alternatives: cosine distance $(1 - \cos(\hat{T}, \hat{S}))$, L_1 distance, and dot product (without normalization). All other components of the framework remain unchanged.

Table XII reports the results on MVTec AD. The squared L_2 distance achieves the highest performance across all metrics (I-AUROC 99.8%, P-AUROC 98.3%, and PRO 94.4%). Cosine distance performs similarly (99.7%, 98.2%, and 94.2%), which is expected since cosine distance is closely related to squared L_2 on normalized vectors. L_1 distance yields slightly lower scores, especially in PRO (93.6%), indicating that it is less sensitive to small but critical discrepancies. Dot product, which omits normalization, performs the worst (98.9%, 96.8%, and 91.2%), confirming that normalization is essential to suppress scale variations. These results justify our choice of squared L_2 in (12).

V. CONCLUSION

This article proposes RGPKD, a KD framework with reconstruction guidance and a prompt module for unsupervised industrial anomaly detection. To overcome limited feature discrepancy in student networks and detail loss during decoding within symmetric distillation structures, we incorporate a reconstruction loss as a strong self-supervised signal and design a prompt module that retrieves and fuses details from the teacher's feature bank. This improves the student's representation of normal features and amplifies its deviation under anomalies. Extensive experiments on benchmarks like MVTec AD show that RGPKD achieves state-of-the-art performance in detection and localization. Future work will investigate more efficient prompt library strategies and extend the framework to domains like video anomaly detection.

REFERENCES

[1] R. Wang, K. Nie, T. Wang, Y. Yang, and B. Long, "Deep learning for anomaly detection," in *Proc. 13th ACM Int. Conf. Web Search Data Mining*, 2020, pp. 894–896.

[2] G. Pang, C. Shen, L. Cao, and A. V. D. Hengel, "Deep learning for anomaly detection: A review," *ACM Comput. Surv.*, vol. 54, no. 2, pp. 1–38, 2022.

[3] L. Ruff et al., "A unifying review of deep and shallow anomaly detection," *Proc. IEEE*, vol. 109, no. 5, pp. 756–795, May 2021.

[4] C. Ding, G. Pang, and C. Shen, "Catching both gray and black swans: Open-set supervised anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, pp. 7388–7398, 2022.

[5] K. Jang, K. E. S. Pilario, N. Lee, I. Moon, and J. Na, "Explainable artificial intelligence for fault diagnosis of industrial processes," *IEEE Trans. Ind. Informat.*, vol. 21, no. 1, pp. 4–11, Jan. 2025.

[6] Y. Qiang, J. Cao, S. Zhou, J. Yang, L. Yu, and B. Liu, "tGARD: Text-guided adversarial reconstruction for industrial anomaly detection," *IEEE Trans. Ind. Informat.*, vol. 21, no. 12, pp. 9297–9308, Dec. 2025.

[7] S. Yoon, Y.-U. Jin, Y.-K. Noh, and F. C. Park, "Energy-based models for anomaly detection: A manifold diffusion recovery approach," in *Proc. 37th Int. Conf. Neural Inf. Process. Syst.*, pp. 49445–49466, 2023.

[8] X. Shen, X. Ge, and W. Wang, "Unsupervised anomaly detection for medical images based on multi-hierarchical feature reconstruction," *IEEE Access*, vol. 12, pp. 151395–151402, 2024.

[9] H. H. Nguyen, C. N. Nguyen, X. T. Dao, Q. T. Duong, D. P. Thi Kim, and M.-T. Pham, "Variational autoencoder for anomaly detection: A comparative study," in *Proc. IEEE Int. Conf. Consum. Electron.*, 2024, pp. 1–6, doi: 10.1109/ICCE56470.2024.10427761.

[10] G. Pang, A. J. Van Den Hengel, C. Shen, and L. Cao, "Toward deep supervised anomaly detection: Reinforcement learning from partially labeled anomaly data," in *Proc. Knowl. Discov. Data Mining*, 2020, pp. 1298–1308.

[11] J. Yang, Y. Shi, and Z. Qi, "DFR: Deep feature reconstruction for unsupervised anomaly segmentation," 2020, *arXiv:2012.07122*.

[12] P. Perera, R. Nallapati, and B. Xiang, "OCGAN: One-class novelty detection using GANs with constrained latent representations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 2893–2901.

[13] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *Comput. Sci.*, vol. 14, no. 7, pp. 38–39, 2015.

[14] C. Peng, Y. Sheng, W. Gui, Z. Tang, and C. Li, "A rolling bearing fault diagnosis method based on multimodal knowledge graph," *IEEE Trans. Ind. Informat.*, vol. 20, no. 11, pp. 13047–13057, Nov. 2024.

[15] H. Deng and X. Li, "Anomaly detection via reverse distillation from one-class embedding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 9737–9746.

[16] Y. Lim et al., "Few-shot semantic segmentation with uncertainty-based joint prototypes," in *Proc. IEEE Int. Conf. Adv. Video Signal-Based Surveill.*, 2025, pp. 1–6.

[17] X. Wang, R. Zhang, C. Shen, T. Kong, and L. Li, "Dense contrastive learning for self-supervised visual pre-training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 3024–3033.

[18] C. Huang et al., "Registration based few-shot anomaly detection," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 303–3019.

[19] H. Cheng, J. Luo, and X. Zhang, "Multimodal industrial anomaly detection via uni-modal and cross-modal fusion," *IEEE Trans. Ind. Informat.*, vol. 21, no. 6, pp. 5000–5010, Jun. 2025.

[20] T. Park, A. A. Efros, R. Zhang, and J.Yan Zhu, "Contrastive learning for unpaired image-to-image translation," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 319–315.

[21] J. Yi and S. Yoon, "Patch SVDD: Patch-level SVDD for anomaly detection and segmentation," in *Proc. Asian Conf. Comput. Vis.*, 2020, pp. 475–489.

[22] H. Guo, C. Zhao, Z. Liu, J. Wang, and H. Lu, "Learning coarse-to-fine structured feature embedding for vehicle re-identification," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 6960–6967.

[23] Y. Jiang, Y. Cao, and W. Shen, "Prototypical learning guided context-aware segmentation network for few-shot anomaly detection. neural networks and learning systems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 7, pp. 12016–12026, Jul. 2025.

[24] K. Roth, L. Pemula, J. Zepeda, B. Schlkopf, and P. Gehler, "Towards total recall in industrial anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 14318–14328.

[25] Z. Liu, Y. Zhou, Y. Xu, and Z. Wang, "SimpleNet: A simple network for image anomaly detection and localization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 20402–20411.

[26] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. 37th Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[27] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 9729–9738, doi: 10.1109/CVPR42600.2020.00975.

[28] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, "MVTec AD—A comprehensive real-world dataset for unsupervised anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 9592–9600.

[29] L. Ruff, R. Vandermeulen, N. Görnitz, L. Deecke, and M. Kloft, "Deep one-class classification," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 4393–4402.

[30] P. Bergmann, X. Jin, D. Sattlegger, and C. Steger, "The MVTec 3D-AD dataset for unsupervised 3D anomaly detection and localization," in *Proc. 17th Int. Joint Conf. Comput. Vis., Imag. Comput. Graph. Theory Appl.*, 2022, pp. 202–213.

[31] S. Lee, S. Lee, and Byung Cheol Song, "CFA: Coupled-hypersphere-based feature adaptation for target-oriented anomaly localization," *IEEE Access*, vo. 10, pp. 78446–78454, 2022.

[32] S. Zhang, X. Wang, and T. Zhang, "Combining multi-scale U-Net with transformer for welding defect detection of oil/gas pipeline," *IEEE Access*, vol. 13, pp. 5437–5445, 2025.

[33] D. Gudovskiy, S. Ishizaka, and K. Kozuka, "CFLOW-AD: Real-time unsupervised anomaly detection with localization via conditional normalizing flows," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 98–107.

[34] V. Zavrtanik, M. Kristan, and D. Skočaj, "DRAEM-A discriminatively trained reconstruction embedding for surface anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 8330–8339.

[35] T. Defard, A. Setkov, A. Loesch, and R. Audigier, "PaDiM: A patch distribution modeling framework for anomaly detection and localization," in *Proc. Int. Conf. Pattern Recognit.*, 2021, pp. 475–489.

[36] Z. Xia, X. Pan, S. Song, Li Erran Li, and G. Huang, "Vision transformer with deformable attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 4794–4803.

[37] C.-F. Chen, Q. Fan, and R. Panda, "CrossViT: Cross-attention multi-scale vision transformer for image classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 357–366.

Kaiyue Wang received the bachelor's degree in software engineering from the School of Computer Science and Technology, Qingdao University, Qingdao, China, in 2024, where she is currently working toward the master of science degree in electronic science and technology.

Her research interest focuses on artificial neural networks.

Chengyan Qin received the Bachelor of Engineering degree in communication engineering in 2025 from the School of Electronic and Information Engineering, Qingdao University, China, where he is currently working toward the master's degree in electronic science and technology.

His research focuses on anomaly detection.

Jieru Chi received the B.S. degree in industrial automation from Shandong University of Technology, Zibo, China, in 1992, the M.S. degree in industrial automation from Shandong University of Technology, Zibo, China, in 1995, and the Ph.D. degree in system theory from Qingdao University, Qingdao, China, in 2019.

She conducted academic research as a Visiting Scholar at Queensland University of Technology, Brisbane, Australia, from 2009 to 2010 and from 2017 to 2018. She is currently a Professor with the College of Electronics and Information, Qingdao University, Qingdao, China. She has authored or coauthored more than 30 papers in journals, such as IEEE TRANSACTIONS ON BIOMEDICAL ENGINEERING. Her research interests include Artificial Intelligence, image processing, and intelligent information processing.

Chenglizhao Chen received the Ph.D. degree from Beihang University, Beijing, China, in 2017.

He is a Professor with the College of Computer Science and Technology, China University of Petroleum (East China), Qingdao, China. He has authored or coauthored more than 50 papers in top-tier journals and conferences. He is currently a Visiting Scholar with Beihang University. His research interests include virtual reality, computer vision, deep learning, and pattern recognition.

Dr. Chen is an Associate Editor for the *Neural Processing Letter* and an Editorial Member of *Electronics*.

Teng Yu received the B.S. degree in communication engineering from the Harbin Institute of Technology, Harbin, China, in 2010, and the Ph.D. degree in electronic and communication engineering from Hanyang University, Seoul, South Korea (under the China Scholarship Council Program), in 2015.

He is a Professor with the College of Electronics and Information, Qingdao University, Qingdao, China. He has authored or coauthored more than 20 SCI papers in the fields of artificial intelligence and computer vision in recent years, and is a Reviewer for renowned journals, such as IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, and EAAI. He research interests include Artificial Intelligence, machine learning, computer vision, and autonomous driving.